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RESEARCH ARTICLE

GNSS-Spoofing Detection via Machine Learning With In-Domain and Cross-Domain Testing and Open-Access I/Q Data

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ABSTRACT Global Navigation Satellite Systems (GNSS) are essential for applications that rely on positioning, velocity, and timing (PVT) information. Unfortunately, a surge in interferences in GNSS frequency bands causes severe security risks for various navigation-based applications. Traditional signal-processing techniques cannot generalize well to different interference types, therefore, Radio Frequency Fingerprinting (RFF) methods using sophisticated Machine Learning (ML) methods are used to capture complex patterns in GNSS signals. However, ML-based spoofing detection studies have relied primarily on the testing of the ML models with the same type of spoofing signals as those used in the training phase (the in-domain testing). This is a strong limitation of ML-based RFF, as real-time spoofers are highly variable and time-varying. Therefore, the ability of an ML algorithm to provide cross-domain generalization across unseen datasets is crucial. To this end, our paper presents a comprehensive analysis of achievable spoofing-detection accuracies using four ML models and compares the in-domain with cross-domain testing accuracies. We used multiple scenarios from two open-access GNSS spoofing repositories: the Texas Spoofing Test Battery (TEXBAT) and the Oakridge Spoofing Test Battery (OAKBAT). For in-domain testing with mixed sample split, all ML models achieve an average classification accuracy greater than 90%. The accuracy for PRN-level becomes slightly lower, with around 80% in GPS L1 and around 75% in Galileo E1. Finally, the average accuracy in the cross-testing approach drops to 66% or less, highlighting the need for more advanced ML models to achieve higher generalization capabilities under complex spoofing conditions.

INDEX TERMS Global navigation satellite systems (GNSS), machine learning (ML), neural networks (NN), OAKBAT and TEXBAT datasets, post-correlation features, spoofing detection.

I. INTRODUCTION

Global Navigation Satellite Systems (GNSS) are responsible for the majority of contemporary PVT applications in outdoor scenarios. Satellite constellations such as GPS, Galileo, BeiDou, Glonass, NavIC, and QZSS, among others, are the basis of PVT services for many navigation applications, such as maritime, land and air transportation, surveying and mapping [1].

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In recent years, interferences in the GNSS bands have increased dramatically, both in terms of jamming [2], [3] and spoofing [4], [5], [6], [7]. GNSS spoofing is the deliberate broadcast of fake GNSS signals that replicate real satellite transmissions, causing the receiver to estimate a false location. Spoofing is stealthier than jamming, as, unlike jamming, which simply blocks or disrupts the signals, the spoofing aims at providing different locations and/or time stamps to a receiver, potentially misleading the receiver into taking actions based on wrong locations. The simplest form of spoofing is the meaconing or repeater attack, where an

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attacker intercepts and rebroadcasts authentic GNSS signals with some delays and typically at a slightly higher power than the original signals; this causes the receiver to calculate an incorrect PVT solution and this can raise important security risks [8], [9]. More sophisticated spoofers also exist, where spoof signals are made more consistent with real GNSS signals in timing, frequency, carrier to noise density ratios (C/N_0) and navigation data and may be transmitted on multiple GNSS frequencies [10], [11]. As Software Defined Radio (SDR) GNSS transmitters are nowadays available in markets, spoofing attacks are occurring with increased frequency [12], despite the fact that the use of wireless spoofers is illegal in many countries, including all European countries.

In order to deal with spoofing in GNSS, there are currently three main approaches in the literature [3], [11]: i) signal-level processing approaches, such as consistency checks at various tracking or navigation levels, signal power monitoring, or geometric techniques based on multi-antenna receivers [3], [13], [14], ii) cryptographic techniques, such as the relatively new Open Service Navigation Message Authentication (OSNMA) feature of Galileo signals [10], [15] and iii) Radio Frequency fingerprinting techniques, typically relying on machine learning to identify hidden patterns or 'fingerprints' in the signals transmitted either from genuine or from spoofing transmitters; these latter techniques are the focus of this paper.

The RFF concept is a broad methodology aiming at detecting the subtle differences between various transmitters, based on their hardware impairments (or 'signatures'). Such hardware impairments can be, for example, I/Q imbalances, phase noise, power amplifier, or Digital-to-Analog Converter (DAC) non-linearities [16], [17], [18]. RFF solutions have typically been implemented via ML algorithms, although few statistical model-based approaches have also been reported so far, e.g., based on Bayesian step-change detectors [19].

Our paper focuses on the spoofing-detection stage, as this is a first step towards further mitigation algorithms. We apply four ML-based approaches, down-selected based on literature searches as described in section II, to detect spoofing presence in GNSS L1/E1 bands. We rely on two well-known open-access repositories with GNSS spoofing data, namely TEXBAT and OAKBAT, described in Section IV.

As the main article contribution, our work presents a comprehensive comparison between ML-based spoofing detection techniques, based on three different testing approaches: in-domain testing with mixed-samples split, in-domain testing with pseudorandom noise sequence (PRN)-level split, and cross-testing approach. Our work is unique not only by addressing these three testing approaches in our methodology but also by using two distinct spoofing repositories with data collected at different sampling rates and under different spoofing conditions, namely TEXBAT and OAKBAT repositories.

The ML models are used together with eight post-correlation features, extracted after processing the I/Q samples via the acquisition and tracking stages. These eight features are: Carrier to Noise density ratio (C/N_0), Doppler shift, Delay Lock Loop (DLL) discriminator output, In-Phase and Quadrature-Phase correlator outputs, Phase Lock Loop (PLL) tracking output, Frequency Lock Loop (FLL) tracking output, and absolute value of the peak correlation at the tracking output. Four widely used ML algorithms, namely **XGBoost, Random Forest, K-Nearest Neighbor (kNN), and Convolutional Neural Network (CNN)** were implemented in our work, based on the studies described in section II. Parts of the work in this paper have been conducted by the first author as a M.Sc. thesis at Tampere University [20].

II. STATE OF THE ART AND MOTIVATION

Given the rise of Artificial Intelligence (AI) in recent years, ML models are widely used for spoofing detection. Decision trees are one of the supervised ML models that rely on multiple weak learners that combine trees to form a strong model. XGBoost is one of these models that has won many ML and data mining challenges [21]. The study in [21] proved that boosting algorithms have low computational complexity, deal well with error correction, and are capable of optimizing model bias in noisy data. Real-time detection of spoofing anomalies in UAV navigation telemetry was also demonstrated using XGBoost [22]. The study claims that even on noisy data from the UAV systems, the model showed robustness, which makes it a strong candidate for immediate spoofing detection environments. Similarly, [23] proposed an XGBoost detection methodology for autonomous vehicles and showed that XGBoost outperformed other models and reached an accuracy of up to 97.64%. This motivated us to pick XGBoost as one of the tested ML algorithms.

Ensemble learning has been widely adopted for classification problems due to its ability to combine multiple learners and improve robustness. In RF fingerprinting, ensemble-based methods have been used to learn discriminative signal characteristics for device and transmitter identification. Guo and Yang proposed a robust RF fingerprint extraction scheme for GNSS spoofing detection, showing that satellite-specific RF fingerprints can help distinguish authentic and spoofed GNSS signals [17]. Similarly, Arun Kumar investigated ensemble learning models for RF fingerprinting of software-defined radios using RF impairments such as IQ imbalance, DC offset, and image leakage. The study reported that Random Forest achieved 85% accuracy at an SNR of 30 dB, while AdaBoost achieved 78%. For 32K multicarrier data, the maximum recognition rates were 92% for Random Forest and 83% for AdaBoost, which shows the effectiveness of ensemble learning for the identification of SDR transmitters [24]. In addition, Galar et al. reviewed ensemble methods for imbalanced classification and highlighted the effectiveness of bagging, boosting, and hybrid-based approaches for improving minority-class detection [25]. These studies support the use of ensemble classifiers, such as Random Forest, as reliable baselines for RF fingerprinting-based spoofing detection.

Another ML model widely used in the literature for detection and classification problems is the k-Nearest Neighbor (KNN), which is a distance-based ML model, its main advantage residing in its low complexity. The research work in [26] used a kNN classifier to detect spoofing and jamming signals, confirming higher accuracy compared to traditional Bayesian-based detection methods. In [27] a signal feature-based spoofing detection approach was proposed for UAV using ML techniques in which the kNN model achieved a detection accuracy of 94.87%, higher than the other models studied. In a similar study [7], using the Carrier-to-noise density ratio and five other Signal Quality Monitoring (SQMs) metrics for model training, kNN again outperformed the other models, achieving 99.55% accuracy.

Although these studies achieved high accuracies on different spoofing recordings, however, none of them studied the classification accuracy of the models in cross-testing scenarios, an approach in which a model is trained on one dataset and tested on different (unseen) datasets.

The authors in [28] also studied Convolutional Neural Networks (CNN) for spoofing detection and achieved a cross-testing accuracy above 90% using pre-correlation spectrogram features from raw I/Q data for training. The results in [28] suggest that when the ML model is trained on a dataset containing complex spoofing patterns, it may generalize well to unseen simplistic spoofers, but the analysis in [28] was limited to very few scenarios and only considered CNN. Our work focuses primarily on post-correlation approaches, able to detect at the PRN level whether a satellite signal comes from a spoofer or a genuine satellite. However, for the sake of completeness, we also include a discussion in Section IV about the distinctions between the pre-correlation methods used in [28] and our post-correlation methods. It should also be noted that post-correlation methods are more widely spread in the RFF literature focused on GNSS than pre-correlation methods, due to their ability to take decisions at PRN level.

This analysis of the previous literature on the detection of GNSS spoofing via ML is also summarized in Table 1 and motivated us to choose the following four ML classes in our analysis: XGBoost, Random Forest, kNN and CNN.

III. GNSS SYSTEM ARCHITECTURE AND SPOOFING MODELS

GNSS is organized into three segments: space segment, ground segment and user segment. The space segment contains satellites in constellations such as GPS, GLONASS, Galileo, and BeiDou that continuously transmit navigation data and signals toward Earth. These satellites operate in orbits between 19,100 and 23,200 kilometres in altitude [31]. The control segment is comprised of ground monitoring, control, and telemetry stations; their tasks are, for example, to monitor the satellite behaviour, to maintain satellite health, and to upload ephemeris and clock corrections [1], [32]. The user segment includes all GNSS receivers used in civil, commercial, and military applications. As many GNSS signals are open, unauthenticated, and received at low powers on Earth, GNSS receivers are exposed to interference threats that degrade the reliability of PVT. Although modern systems such as Galileo OSNMA introduce some authentication features, most GNSS signals still remain vulnerable to environmental effects and intentional and unintentional interferences [33].

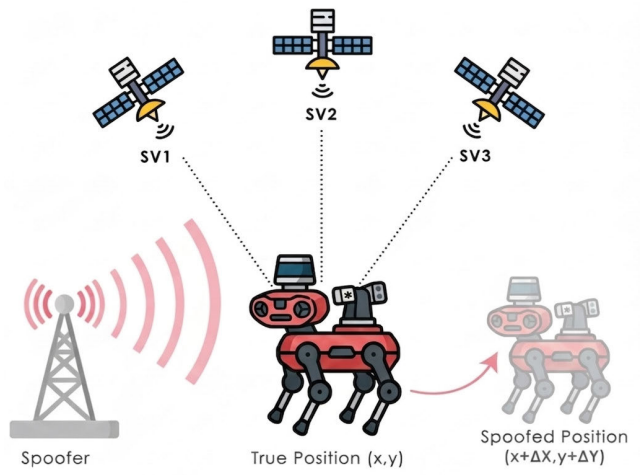

FIGURE 1. Illustration of the change in position due to a spoofer attack on a robot with GNSS receiver.

Spoofing, addressed here, is a major risk to the integrity of GNSS-based systems [4]. As a result of the spoofing attack, the receiver may compute an incorrect position, as shown in Fig. 1. The authors in [34] classify spoofing as simple, intermediate, or sophisticated based on the complexity of the attacks. Simple attacks can be detected easily by noticing sudden position jumps or signal anomalies. Intermediate spoofing may be implemented via a pull-off approach, where the forged signal remains closely aligned with the authentic signal and progressively captures the tracking loops without raising alarms. Sophisticated spoofing constitutes the most advanced form and is particularly difficult to detect, especially when transmissions originate from multiple locations [35].

A simplified mathematical model of the authentic $y(t)$ and spoofed $y_s(t)$ GNSS signals is presented below, based on [35]:

$$y(t) = \mathrm{Re}\left\{ \sum_{k=1}^{N} a_k\, D_k[t-\tau_k(t)]\, C_k[t-\tau_k(t)] \times e^{\mathrm{j}(\omega_c t - \phi_k(t))} \right\}. \quad (1)$$

Equation (1) represents the authentic GNSS signal. Each satellite k transmits its own signal, where C_k represents the spreading code, a_k denotes the signal amplitude, D_k represents the navigation data bits, ω_c is the carrier angular frequency, $\phi_k(t)$ is the carrier phase, and $\tau_k(t)$ is the code delay. The operator Re extracts the real part of the

TABLE 1. Summary of prior ML-based spoofing detection work on TEXBAT, OAKBAT, and other datasets.

Algorithm	Datasets	Features	Train-Test Split	Reported In-Domain Testing Accuracy (%)	Cross-Domain Testing	Reference
XGBoost	TEXBAT	C/N_0, Doppler	Not reported	97.37	Not reported	[27]
	TEXBAT (DS2–DS7)	C/N_0, Signal power, 5 SQMs	75/25	98.64	Not reported	[7]
	AV-GPS DS1	Flight parameters	10 folds	97.6	Not reported	[23]
Random Forest	TEXBAT	C/N_0, Doppler	Not reported	91.99	Not reported	[27]
	TEXBAT (DS2–DS7)	C/N_0, Signal power, 5 SQMs	75/25	99.5	Not reported	[7]
	AV-GPS DS1	Flight parameters	10 folds	97.1	Not reported	[23]
kNN	TEXBAT (DS3, DS8)	C/N_0, DLL (E,P,L)	80/20	94.5	**Reported-** Train: PRN19 (DS8), Test: other PRNs from other DS sets; average accuracy only 47.6% with PRN-level identification	[29]
	TEXBAT (DS2–DS7)	C/N_0, Signal power, 5 SQMs	75/25	99.55	Not reported	[7]
	AV-GPS DS1	Flight parameters	10 folds	94.91	Not reported	[23]
CNN	TEXBAT (DS2–DS4, DS8), OAKBAT (OS2)	I/Q magnitude spectrogram as feature	80/20	≈ 90	**Reported**, average 84.94% with incomplete TEXBAT datasets (4 sets) and without PRN-level identification	[28]
	TEXBAT and Simulated Data	Images of I/Q correlation values	Train:80, Val:10, Test:10	96.04	Not reported	[30]
XGBoost, Random Forest, kNN, CNN	TEXBAT (DS1-DS8) and OAKBAT (OS10-OS13)	Eight post-correlation features	80/20, 50/50	Tables 6, 7 and Fig. 8; accuracies up to 99.5% with mixed-samples split and up to 93% with PRN-level split.	**Reported**, average between 65.38% (CNN) and 68.48% (XGboost) with all TEXBAT and 4 OAKBAT scenarios and with PRN-level identification.	**Our work**

complex-valued expression.

$$y_s(t) = \mathrm{Re}\left\{ \sum_{k=1}^{N_s} a_{s_k} \tilde{D}_k[t - \tau_{s_k}(t)]\, C_k[t - \tau_{s_k}(t)] \times e^{\mathrm{j}(\omega_c t - \phi_{s_k}(t))} \right\}, \quad k = 1, \ldots, N_s. \tag{2}$$

A spoofer generates a counterfeit signal by reproducing the same structural components while modifying key parameters such as the signal amplitude, code delay, and carrier phase. Equation (2) describes this spoofed signal model, where the subscript s_k denotes spoofed-signal quantities. By carefully adjusting these parameters, the spoofer ensures that the counterfeit signal closely tracks the authentic signal, facilitating a seamless takeover of the receiver's tracking loops.

$$y_{\mathrm{tot}}(t) = y(t) + y_s(t) + \eta(t). \tag{3}$$

In a spoofing condition, the total signal received can be expressed as shown in eq. (3). This signal consists of elements of both the genuine signal, the spoofed signal, and the added noise component of the signal, denoted here by $\eta(t)$. This compact model shows the effect of spoofing upon the signal at the receiver's front end, which provides the basis for conducting spoofing signal analysis and spoofing detection.

Modern GNSS literature provides various spoofing detection techniques that can be broadly classified into non-learning and learning-based detection techniques. Non-learning methods also use correlation peak monitoring [26], [36], [37], signal power [38], antenna array processing [39], data-bit techniques [40], and pseudorange measurements [41]. Learning based methods utilize machine learning and deep learning models to classify clean and spoofed samples in GNSS data using a variety of publicly available datasets [18], [33], [42]. A detailed review of anti-spoofing techniques is also presented in [38].

The non-learning methods discussed here rely on manually designed thresholds, due to which they fail in sophisticated attacks. Therefore, ML and DL models are used which can even extract subtle patterns in complex spoofing scenarios

leveraging variety of features. However, most ML based studies rely merely on simple mixed-sample splits to train the model, as shown in Table 1. These experiments yield higher accuracy than their PRN-level counterparts, but they ignore the PRN-level accuracies (i.e. identifying which PRNs are spoofed among all visible genuine and fake satellites), which can be more realistic for operational deployment. To overcome these gaps, the present study conducts systematic in-domain and cross-domain testing experiments using TEXBAT and OAKBAT spoofing datasets and addresses both the mixed-samples split (for the in-domain testing) and PRN-level split (for both in-domain and cross-domain testing). The methodology described in Section IV outlines the data preparation, feature extraction, and model development pipeline adopted in this work.

IV. METHODOLOGY

A. PROCESSING FLOWCHART AND DESCRIPTION OF THE USED DATASETS

The proposed framework is illustrated in Fig. 2 where the raw GNSS signal datasets from TEXBAT [43] and OAKBAT [44] were processed using FGI-GSRx MATLAB software [13] for the effective extraction of post-correlation features. The TEXBAT and OAKBAT datasets provide a rich amount of clean and spoofed GNSS signal scenarios and have been used by many researches in their spoofing detection studies, as shown in Table 1. Nevertheless, none of the works so far, to the best of the Authors' knowledge, has systematically investigated both TEXBAT and OAKBAT datasets.

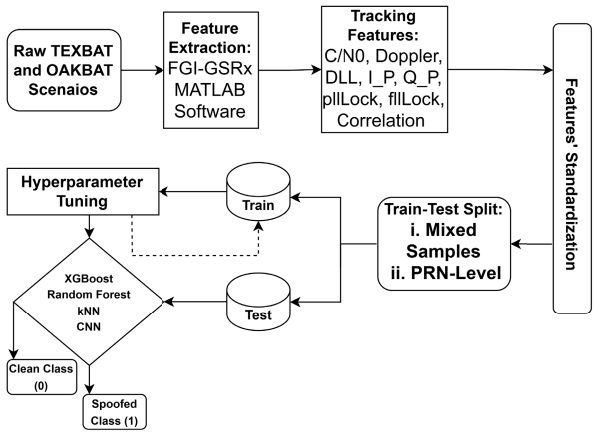

FIGURE 2. Flowchart illustrating the methodological workflow of spoofing detection in GNSS datasets.

The TEXBAT and OAKBAT scenarios are recorded in 32-bit I/Q format (meaning 16-bit quantization for I and 16-bit quantization for Q channels). The OAKBAT GPS L1 and Galileo E1 scenarios match the TEXBAT GPS L1 scenarios in most respects, except the sampling rates, as summarized in Table 2. However, OAKBAT does not provide scenarios corresponding to TEXBAT DS7 and DS8. The rows in Table 2 make the GPS-to-Galileo correspondence explicit. The signal-processing parameters in Table 3 are applied to parse the binary files and extract features from both datasets. The only main difference lies in the sampling rate: TEXBAT was recorded at 25 MHz, while OAKBAT was recorded at 5 MHz [44]. Since these datasets were created specifically to evaluate spoofing detection approaches, they are a strong fit for this study.

B. EXTRACTED FEATURES

The FGI-GSRx open-access software receiver, implemented in MATLAB [13], was used to extract post-correlation (i.e., tracking-level) features, including carrier-to-noise density ratio (C/N_0), Doppler, Delay Lock Loop (DLL) outputs, in-phase (I_P) and quadrature (Q_P) prompt components, PLL lock indicator, FLL lock indicator, and the in-prompt correlation values. These features were chosen because of their relevance to detect spoofing behavior, based on the literature overview described in Section II. The features were arranged over equal-length windows (100 s of clean signal and 100 s of spoofed signal) and aligned per PRN identified during the acquisition stage. To evaluate performance on TEXBAT and OAKBAT datasets, three different train–test approaches were defined. In all cases, the task was treated as binary classification (clean = 0, spoofed = 1). Because the features have different magnitudes and units, they were standardized according to Eq. (4) using **StandardScaler** from the sklearn.preprocessing library.

$$z = \frac{x - \mu}{\sigma}, \tag{4}$$

where z represents the standardized score of sample x, μ and σ represents the mean and standard deviation of the training. Each time the model was trained, the data were shuffled, so the model learns general patterns. For XGBoost, Random Forest and kNN, **five** random training–testing runs were performed. For CNN, **ten** runs were performed because its performance showed slightly higher variability in the test set. Accuracy is reported as the mean ± standard deviation in multiple runs. The hardware/software environment for model training and evaluation is summarized in Table 4.

C. ML ALGORITHMS HYPERPARAMETERS

The hyperparameters of the models were tuned using a randomized search cross-validation (CV) with a three-fold cross-validation and five sample configurations or iterations. The randomized search was selected instead of a grid-search cross-validation to enable efficient exploration of a high-dimensional parameter space at reduced computational cost while retaining competitive performance. We provide the combinations of hyperparameters in Table 5 chosen for the training that gave the highest cross-validation accuracy.

The hyperparameter ranges in Table 5 for Random Forest, XGBoost, and kNN were selected based on the performance in the validation data. The number of estimators controls the number of decision trees in the Random Forest and XGBoost

TABLE 2. Scenario mapping between the TEXBAT and OAKBAT datasets [43].

TEXBAT (GPS L1)	OAKBAT (GPS L1)	OAKBAT(Galileo E1)	Attack and Receiver state	Power Advantage (dB)
CleanStatic	CleanStatic	CleanStatic	No spoofing (Static)	None
CleanDynamic	CleanDynamic	CleanDynamic	No spoofing (Dynamic)	None
DS1	OS1a	OS9a	Static switch (Static)	None
DS2	OS2	OS10	Overpowered time push (Static)	10
DS3	OS3	OS11	Matched-power time push (Static)	1.3
DS4	OS4	OS12	Matched-power position push (Static)	0.4
DS5	OS5	OS13	Overpowered time push (Dynamic)	9.9
DS6	OS6	OS14	Matched-power position push (Dynamic)	0.8
DS7	N/A	N/A	Matched-power time push (Static)	Matched
DS8	N/A	N/A	Security Code Estimation and Replay (SCER) attack (Static)	Matched

TABLE 3. Recording and parsing settings for the TEXBAT GPS L1 I/Q data [43] and OAKBAT Galileo E1 I/Q data.

Parameter	Value / notes
I/Q sample format	32-bit complex sample
Absolute timestamp	25.03 s (recording start time)
Relative timestamp	−2.61 s (relative offset)
Sampling interval	40 ns (time between consecutive samples)
I/Q sampling rate	25 MHz for TEXBAT and 5 MHz for OAKBAT
Bandwidth	20 MHz (effective processed bandwidth)
Center frequency	1575.42 MHz
Gain	8.74 (applied gain factor)

TABLE 4. Hardware/software environment for the model training and evaluation.

Parameter	Value
Hardware platform	Lenovo ThinkPad
CPU	Intel Core i5 (11th Gen), 2.60 GHz
Memory	16 GB RAM, 1 TB SSD
Operating system	Windows 11 (23H2, 64-bit)
MATLAB	R2024b
Python / IDE	Python 3.11 (PyCharm Community 2024.1.4)

TABLE 5. Hyperparameter search space used for model tuning (randomized-search cross-validation) and the selected CNN configuration.

Method	Search space / configuration
Random Forest	Estimators: 30–250; Depth: 3–12; Min split: 3–12; Min leaf: 1–6.
XGBoost	Estimators: 30–250; Depth: 3–15; Learning rate: 0.03–0.30.
kNN	k: 1–15; Weighting: uniform vs. distance; Metric power: $p = 1$ (Manhattan) or $p = 2$ (Euclidean).
CNN	Conv1D filters: (64, 128); Kernel length: 3; Dropout: 0.4; Optimizer: Adam; Batch: 128; Epoch cap: ≤ 60 with early stopping.

models, the depth parameter manages complexity, and the min split controls the sensitivity of a tree during splitting. The learning rate corrects the loss caused by the previous decision tree in the boosting algorithms. The parameter k controls the number of neighbors in the kNN model to make a prediction. In addition, weights define the influence of each neighbor on the prediction based on the distance from the query point.

The CNN hyperparameters in Table 5 were empirically tuned using the validation set. The network took samples in the shape or size of (100, 8). The best-performing model on validation data had 64 neurons in the first Conv1D layer, 128 neurons in the second, followed by a fully connected Dense(64) layer and a Sigmoid layer for binary output. Adam optimizer was used to optimize the network due to its adaptive step-size updates. Each hidden layer also used Relu as an activation function. The network also used Binary Cross Entropy (BCE) as a loss function. Training was capped at 60 epochs with early stoppage to avoid overfitting and address differences in convergence time across different datasets.

The ML models from Table 5 relied on widely used Python libraries such as NumPy, pandas, SciPy, matplotlib and seaborn. kNN and Random Forest were implemented with scikit-learn, and XGBoost was implemented through the xgboost library in Python. CNN was built and trained using TensorFlow and Keras. The evaluation of the test data involved the calculation of **confusion matrices** and the **plotting of generalization curves** for each train-test set. Figure 3 illustrates train-test error curves for XGBoost, Random Forest and CNN when TEXBAT DS7 PRNs were used for training and DS8 PRNs for testing. Generalization

curves represent the relationship between training error and test/validation error, and are used to evaluate whether a model is underfitting, overfitting, or generalizing well to unseen data. A small gap between the two errors indicates better generalization, whereas a large gap suggests overfitting. High training and test errors indicate underfitting.

D. BRIEF DISCUSSION ON POST-CORRELATION VERSUS PRE-CORRELATION APPROACHES

Pre-correlation RFF methods operate directly on the raw GNSS IF/baseband samples, preserving detailed hardware-originated features such as oscillator instabilities, phase noise, power-amplifier non-linearities, and IQ imbalance. Because these low-level impairments remain intact prior to any correlation processing, pre-correlation approaches typically provide much higher accuracy than post-correlation approaches [45]. However, the pre-correlation approaches suffer of high computational burden (e.g., the need to store large spectrograms of input data instead of only few post-correlation features) and are unable to detect the spoofing signal at PRN level.

Unlike pre-correlation methods, the post-correlation RFF techniques rely on higher-level receiver output, such as the features described in Section IV-B. This greatly reduces computational load and enables deployment in mass-market GNSS devices, as well as PRN-level detection or classification. This ability to classify a spoofing signal at the PRN level motivated us to focus in this paper on post-correlation approaches. However, for the sake of completeness and because we have previously analyzed the cross-domain testing capabilities of a CNN algorithm with few dataset combinations from TEXBAT [28], we are also showing in Fig. 4 the confusion matrix based on cross-domain testing with pre-correlation data (via CNN and magnitude of the spectrogram) based on all TEXBAT datasets and on five OAKBAT datasets.

The combinations tested in [28] were DS2-DS3, DS2-DS4, DS3-DS2, DS3-DS4, DS4-DS2, DS8-DS2, DS8-DS3, and DS8-DS4, where the first name in DSx-DSy notation tells the name of the training dataset (DSx) and the second name (DSy) tells the name of the testing dataset. The average cross-domain testing accuracy with pre-correlation data reported in [28] over these nine combinations was 83.83%. In Fig. 4 (top), we now tested all 64 possible combinations of TEXBAT datasets, and we obtained an average accuracy calculated over all of these equal to 69.09%. It should be noted that when we only look at the average accuracy on the same nine cross-domain testing combinations used in [28], we obtain 82.75%, which is very similar to [28]. The 1% difference can be explained by a slightly smaller processing window here of 50s (instead of 60s used in [28]), chosen to speed up the computations. Similarly, the average cross-domain testing accuracy with pre-correlation OAKBAT data is 72.45%, while in-domain testing with OAKBAT data reaches 100% accuracy.

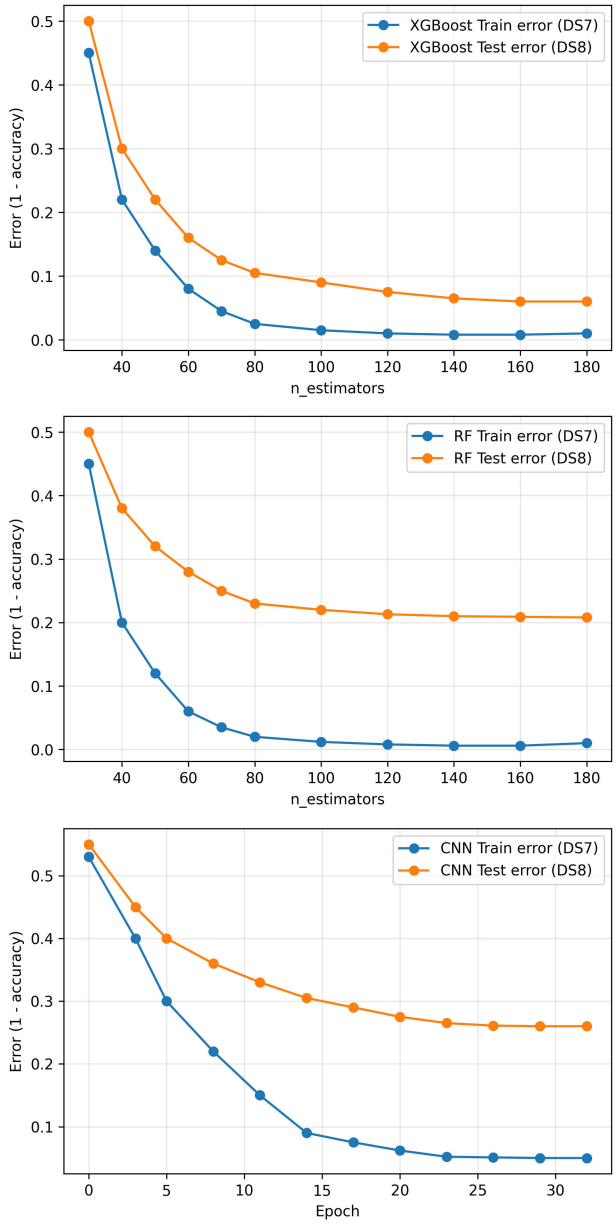

FIGURE 3. Generalization behaviour of the ML models used in PRN-level cross-testing. Top: XGBoost; middle:Random Forest, bottom: CNN.

The take-away point from the discussion in this subsection is that, while pre-correlation approaches have previously been reported to give very high accuracies for GNSS spoofing detection, the results in the literature so far have investigated only a subset of TEXBAT datasets (and none from OAKBAT datasets) and failed to capture all possible cross-domain testing combinations. When one looks at more combinations between training and testing datasets, the pre-correlation approaches do suffer from lack of generalization ability. Moreover, pre-correlation approaches cannot distinguish whether a particular PRN is spoofed or not, as the PRN information is not available before the tracking stage. Therefore, in what follows, we will concentrate on the post-correlation approaches.

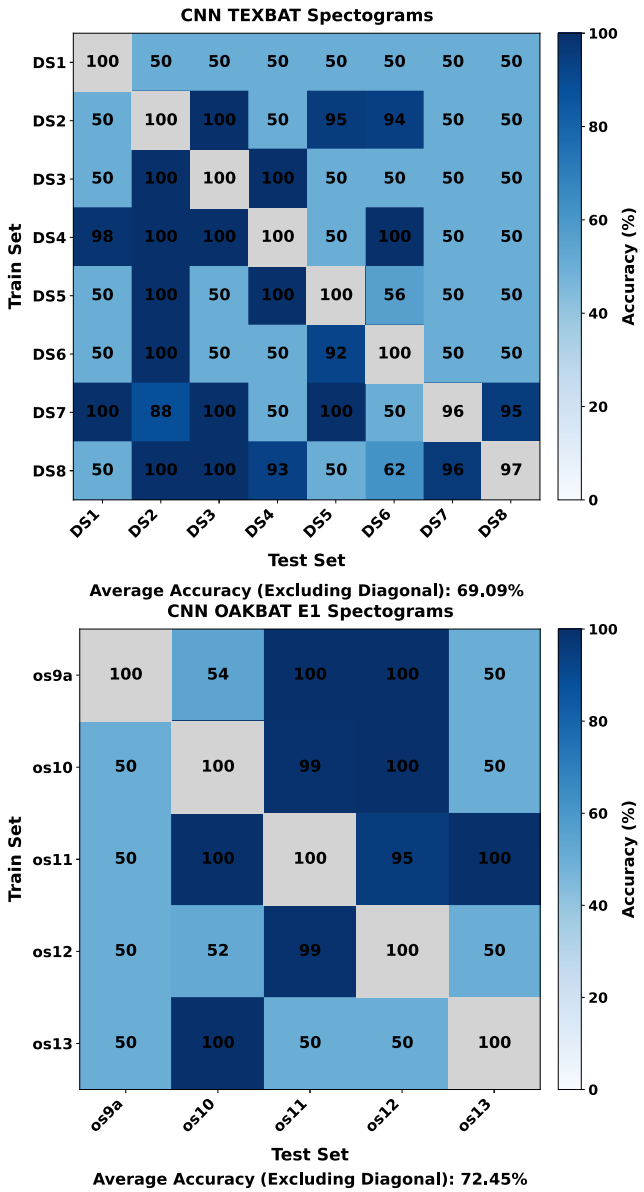

FIGURE 4. Cross-domain testing performance based on a pre-correlation approach (CNN with magnitude spectrogram as a feature). **Top:** TEXBAT GPS L1; **bottom:** OAKBAT Galileo E1.

V. RESULTS

This section analyzes the experimental results under three classification scenarios: in-domain testing with mixed-samples split, in-domain testing with PRN-level split, and cross-domain testing with PRN-level split (as a mixed-samples split makes no sense in cross-domain testing). Each approach is described and results are illustrated in this section.

A. IN-DOMAIN TESTING

1) IN-DOMAIN TESTING WITH MIXED-SAMPLES SPLIT

In this approach, clean and spoofed samples were stored in separate directories, where each CSV file represented an individual PRN and contained its corresponding tracking features. Each file has a size of [50000 (rows), 8 (feature columns)] where each row corresponds to 1 ms of processed data. Clean and spoofed data were combined, after being labelled, and then distributed in the 60:20:20 training, validation, and testing ratios. Similar approaches with varying train-test splits are being employed by [7], [28], [29], and [42]. While the data are saved as one.csv file per visible PRN, the processing in the mixed-samples approach is done over all PRNs at a time, without aiming to distinguish whether a particular PRN is spoofed or clean, but only whether the sample drawn from the pool of PRNs is spoofed or clean.

Table 6 summarizes the results for the TEXBAT and OAKBAT datasets using this mixed-samples approach. In general, XGBoost, Random Forest, kNN, and CNN performed strongly in their ability to classify whether sample belongs to the spoofed or clean pools, with an average accuracy of each model greater than 95± std % in the TEXBAT GPS L1 case, with mean over all eight considered TEXBAT scenarios. In the OAKBAT Galileo E1 scenarios, the models achieved a mean classification accuracy greater than 92± std %, with a mean taken over the four Oakbat scenarios considered. As shown in Table 6, the XGBoost model achieved a near perfect average accuracy greater than 99 ± std % in the GPS L1 and Galileo E1 scenarios.

Therefore, for the in-domain testing with the mixed-samples approach, the XGBoost-based classification outperformed the other three tested ML algorithms and reached average accuracies over the TEXBAT and OAKBAT scenarios greater than 99%, which are comparable or better than the other results reported in the in-domain testing literature so far, when compared with Table 1. A drawback of this mixed-sample approach is that it does not look at PRN-level accuracies and does not try to classify each visible PRN. Such a method is therefore suitable only when all satellites on sky are spoofed, but fails to identify the likelihood of a particular PRN in the mixture being spoofed or genuine. This is why our next sub-sections also look at the in-domain testing with PRN-level split approach.

2) IN-DOMAIN TESTING WITH PRN-LEVEL SPLIT

In an in-domain testing with PRN-level split approach, the classifier aims to classify each individual PRN (from a satellite identified as visible after the tracking loop) to determine whether it is a clean or a spoofed PRN. The accuracy of the PRN-level splits is computed by taking the average of percentage of PRNs in clean and spoofed classes. This in-domain testing with PRN-level split trains each ML model on a subset of PRNs labeled as clean or spoofed, and tests the trained model on separate PRNs and time windows to assess its generalization. However, in the in-domain testing, all these samples are still coming from the same datasets; e.g., if training was done on TEXBAT DS7, the testing is still done on DS7, but different PRNs and different time windows, in such a way that the test PRNs are totally disjoint from the training ones. The purpose of this split is twofold: (a) to assess the temporal generalization of the ML models within a single

TABLE 6. Classification accuracy of ML models on the TEXBAT and OAKBAT datasets in mixed-samples split.

TEXBAT GPS L1									
Classifier	DS1	DS2	DS3	DS4	DS5	DS6	DS7	DS8	**Avg (%)**
XGBoost	100±0.3	100±0	100±0.2	100±0.1	100±0	99±0.4	98±1.6	99±0.2	**99.5±0.3**
Random Forest	98±1.6	100±0.2	98±1.1	98±0.7	99±0.9	97±1.3	94±2.4	97±2.3	**97.6±1.3**
kNN	98±1.4	100±0.6	96±2.1	98±0.5	99±0.5	95±2.4	79±3.1	97±1.7	**95±1.5**
CNN	97±0.8	100±0	98±1.8	97±1.5	99±0.6	98±1.7	96±3.1	99±0.3	**98±1.2**
OAKBAT GPS L1									
Classifier	OS1	OS3	OS4	OS5	**Avg (%)**				
XGBoost	97±2.4	100±0	100±0.3	100±0.3	**99±0.7**				
Random Forest	76±4.1	97±0.5	100±1.7	100±0	**92±1.5**				
kNN	77±2.9	96±0.9	98±1.3	100±0	**92±1.7**				
CNN	89±3.2	96±1.6	97±1.9	97±3.3	**96±1.7**				
OAKBAT Galileo E1 Scenarios									
Classifier	OS10	OS11	OS12	OS13	**Avg (%)**				
XGBoost	100±0	100±0	98±1.1	100±0.3	**99.5±0.3**				
Random Forest	100±0.2	100±0	96±2	100±0.1	**98.7±0.7**				
kNN	98±0.8	98±1.4	89±4.3	97±1.9	**95.5±1.6**				
CNN	99±0.8	99±0.2	89±4.3	97±1.9	**96.6±1.6**				

scenario and (b) to establish a baseline for interpreting the results of stricter cross-scenario testing.

Table 7 presents the results of the in-domain evaluation obtained with a PRN-level split. In the TEXBAT GPS L1 scenarios, all the models compared showed comparable accuracy (approximately 90 ± std %), suggesting reliable detection of spoofing behavior within the same scenario, but slightly smaller accuracies than those obtained with mixed-sample split from Table 6. For OAKBAT scenarios, the accuracies are also smaller for PRN-level split than for mixed-samples split, but the best performance is obtained with CNN for OAKBAT GPS L1 signals and with XGboost for OAKBAT Galileo signals. Given that XGBoost performance is either the best or within less than 6% from the best one, the XGBoost classifier also offers a good tradeoff between performance and complexity for in-domain testing with PRN-level split. The complexity considerations are addressed next.

3) COMPLEXITY CONSIDERATIONS FOR THE IN-DOMAIN TESTING

Table 8 shows the complexity comparison for the four ML algorithms considered. We show the average computation times for the Hyperparameter Optimization (HPO) stage, the training stage and the testing stage. The computation times are shown via their mean ± their standard deviation, expressed in seconds. For simplicity and because we got similar results for the other datasets, we illustrate these complexity values only for TEXBAT DS7 dataset. We see that XGBoost also outperforms the other models in terms of complexity, as it takes less time to search for hyperparameters, train, and test the data than the other three ML models considered.

B. CROSS-DOMAIN TESTING

In cross-domain testing, the training is done on one dataset at a time and tested with a different dataset; therefore, the PRNs are different between the training and testing stages, so we can only talk about a "cross-domain testing with PRN-level split"; there is no mixed-sample option in this case. An illustrative example is shown in Fig. 5 for all visible PRNs from TEXBAT DS6 used in training and TEXBAT DS5 used in testing. With this cross-domain testing approach, similar to the in-domain testing approach with PRN-level split, we are able to classify each PRN into a clean or a spoofing class. The next results will show the averages over all visible PRNs in order to present all tested combinations in a compact form. However, Fig. 5 reminds us that the actual detection can be performed at PRN level in this approach.

Figures 6 and 7 show examples of the average detection probabilities in cross-domain testing, for different combinations of training and testing datasets, and for three of the four ML algorithms considered. kNN was dropped from cross-domain testing, as it gave the poorest results among the algorithms considered in the in-domain testing and higher training time. Figure 6 is for all 64 combinations of TEXBAT datasets (56 cross-domain and 8 in-domain), while Fig. 7 is for 16 combinations of OAKBAT datasets (12 cross-domain and 4 in-domain).

Figure 8 shows a more compact illustration of all the results so far: the top bars show the in-domain with mixed samples, the middle show the in-domain with PRN-level split, and the bottom graphs show the cross-domain testing results, by averaging the results over all possible cross-testing combinations (56 for TEXBAT and 12 for OAKBAT). The results in Fig. 8 indicate that the performance in cross-domain testing deteriorated with about 30% compared to in-domain

TABLE 7. Classification accuracy of ML models using TEXBAT and OAKBAT datasets in the in-domain PRN-level split.

TEXBAT GPS L1									
Classifier	DS1	DS2	DS3	DS4	DS5	DS6	DS7	DS8	**Avg (%)**
XGBoost	100±0	100±0	87±1.7	87±1.3	96±0.7	77±2.9	78±2.2	91±0.6	**89.5±1.1**
Random Forest	95±0.9	100±0	92±1.2	84±1.8	94±0.7	72±2.1	98±0.8	92±0.6	**90.8±1.0**
kNN	98±1.8	100±0	90±2.7	85±3.4	93±1.2	72±1.9	71±4.3	97±1.7	**88.2±2.1**
1D-CNN	100±0	100±0	97±0.4	94±1.3	97±0.6	79±1.8	55±3.1	98±0.8	**90±1.0**
OAKBAT GPS L1									
Classifier	OS1	OS3	OS4	OS5	**Avg (%)**				
XGBoost	64±3.2	97±2.0	88±2.1	99±0.9	**87±1.6**				
Random Forest	63±2.9	98±1.0	91±0.6	98±0.3	**87±1.2**				
kNN	58±4.6	82±3.0	87±2.2	98±1.7	**81±2.8**				
1D-CNN	81.7±2.4	94±1.0	96.5±3.0	99.7±1.0	**93±1.8**				
OAKBAT Galileo E1 Scenarios									
Classifier	OS10	OS11	OS12	OS13	**Avg (%)**				
XGBoost	89±0.6	76±1.0	62±0.3	89±2.5	**78.7±1.1**				
Random Forest	91±0.5	74±1.2	62±1.7	85±0.9	**78±1.0**				
kNN	92±1.3	78±2.1	58±3.2	84±2.8	**78±2.35**				
1D-CNN	91±1.9	77±0.5	63±2.5	73±1.5	**75.9±1.6**				

TABLE 8. Average computation time (mean ± std, in seconds) of the models on TEXBAT DS7.

Mixed Samples Split (TEXBAT DS7)			
Model	**HPO**	**Train**	**Test**
XGBoost	87.25 ± 6.25	11.55 ± 2.52	0.37 ± 0.09
Random Forest	782 ± 16.4	251.7 ± 9.3	9.37 ± 2.16
kNN	254.6 ± 33.8	4.30 ± 0.89	91.18 ± 24.37
CNN	N/A	74.87 ± 1.94	0.43 ± 0.07
PRN-Level Split (TEXBAT DS7)			
Model	**HPO**	**Train**	**Test**
XGBoost	50.10 ± 12.36	6.28 ± 1.37	9.13 ± 0.34
Random Forest	584 ± 10.6	134.6 ± 4.3	12.75 ± 1.89
kNN	184.5 ± 4.8	2.21 ± 0.36	220 ± 10.1
CNN	N/A	13.72 ± 0.74	0.63 ± 0.06

FIGURE 5. PRN-wise performance of XGBoost model trained with TEXBAT DS6 and tested on DS5. **Top:** clean PRNs detection accuracies; **bottom:** spoofed PRNs detection accuracies.

testing. The XGBoost algorithm is still the algorithm with the best performance among the considered ones, but the average cross-domain testing performance with these classical ML algorithms is only 68.5% over all considered TEXBAT and OAKBAT datasets, compared to 99.3% and 85.1% for in-domain testing with mixed-samples split and PRN-level split, respectively.

This pattern suggests that the classical ML models, such as those considered here, capture scenario-specific characteristics effectively, but their performance decreases when transferred to unseen scenarios, indicating limited cross-scenario generalization. DS7 in TEXBAT is considered the hardest spoofing scenario containing subtle spoofing patterns, so when XGBoost was trained on DS7 and tested on others, the model showed good classification accuracy (e.g., see Fig. 6). However, training on simpler spoofing scenarios which have higher power advantages such as DS2 yielded the lowest accuracy on subtle or more complex

spoofing scenarios. Therefore, this aspect should be considered while designing the receiver so that the model trained with more rigid spoofing conditions can be effective in accurately classifying clean and spoofed samples.

Some scenarios also showed the counter-intuitive effect that cross-domain testing outperforms the in-domain testing, e.g., TEXBAT DS7-DS8. Although DS7 and DS8 share similar spoofing characteristics, DS8 uses a zero-delay Security Code Estimation and Replay (SCER) attack with real-time navigation data, yielding a more stable and predictable spoofing signature than DS7. Therefore, training on DS7's noisier behavior can generalize to DS8's more deterministic condition. Each scenario has different levels of distortion, spoofing power, multipath, and noise imbalance, which can reduce in-domain accuracy and make some cross-domain testing may be easier than in-domain testing.

C. DISCUSSION

The following key take-aways can be drawn from the results.

- **The split strategy matters:** the mixed-sample split is optimistic due to shared PRNs and a larger training fraction. When switching to an in-domain PRN-level split, the mean accuracy decreases across TEXBAT and OAKBAT due to unseen PRNs, making the task harder. Despite the lower accuracy, the PRN-level splits provide more useful insights than mixed-samples split about the generalization capabilities of the ML models in real situations where unseen PRNs or new signal patterns occur at test time.
- **The cross-domain testing is the real stress test:** In addition, cross-domain evaluations resulted in about 30% lower mean accuracies than in-domain testing. The primary reason is that the TEXBAT and OAKBAT scenarios differ in collection environment (static or dynamic), spoofing approach (matched-power, overpowered, replay) and signal characteristics, as summarized in Table 2. However, the models that perform well in cross-domain testing are stronger candidates for practical use, including deployment in commercial GNSS receivers exposed to diverse spoofing environments.
- **Accuracy is highly dependent on the spoofing scenario:** TEXBAT DS7 is widely regarded as one of the most challenging scenarios for spoofing detection and consequently produced lower classification performance. DS7 was particularly challenging: kNN reached $79 \pm 3.1\%$ in Table 6 (most others$>$ 94%), and CNN dropped to $55 \pm 3.1\%$ under the in-domain PRN-level split. Similarly, within OAKBAT, OS1 (GPS L1) and OS12 (Galileo E1) behaved as the most challenging scenarios and lowered the mean accuracies of the models. Similarly, comparing GPS with Galileo constellations, TEXBAT GPS L1 was generally the easiest, OAKBAT GPS L1 was intermediate, and OAKBAT Galileo E1 was the most difficult constellation.
- **Model selection matters:** The results further highlight that selecting an appropriate ML model is critical for spoofing detection. In general, kNN was the weakest model in our experiments. XGBoost maintained the most consistent accuracy in all three splitting strategies,

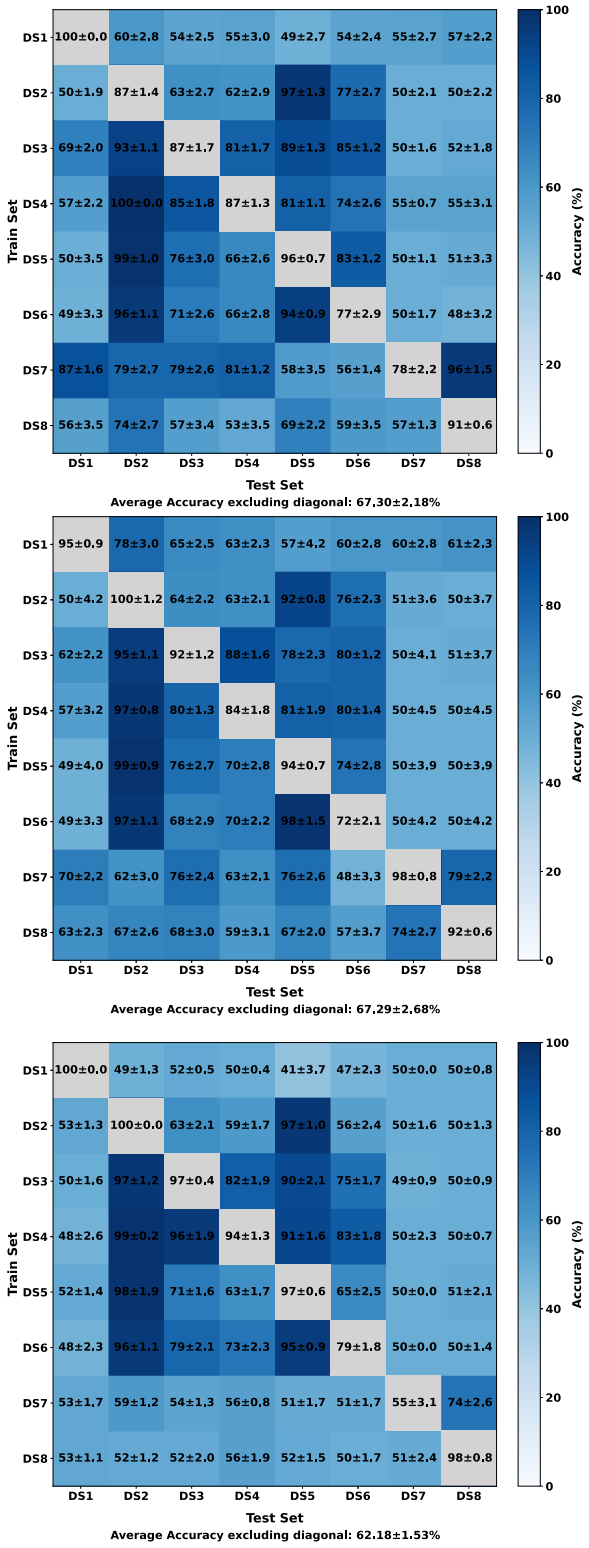

FIGURE 6. Cross-testing performance on the TEXBAT GPS L1 dataset. **Top:** XGBoost; **middle:** Random Forest; **bottom:** CNN.

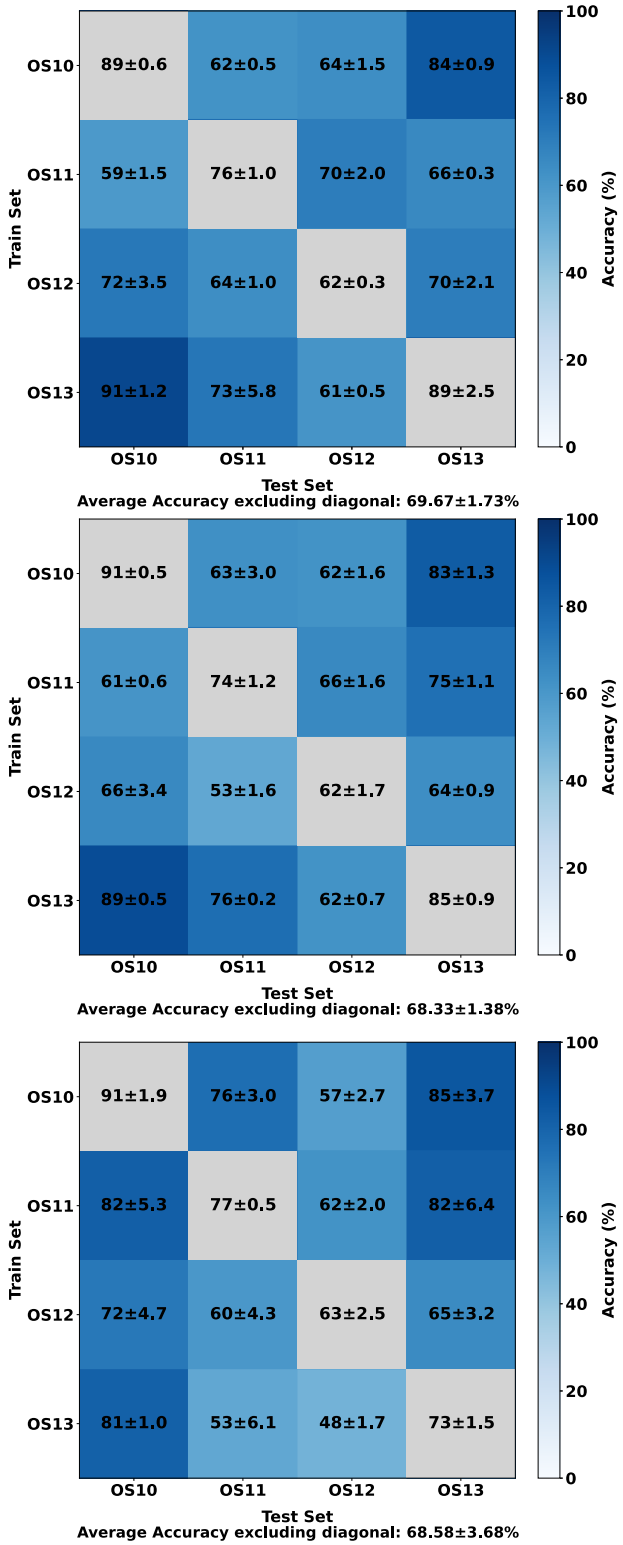

FIGURE 7. Cross-testing performance on OAKBAT Galileo E1 dataset. **Top:** XGBoost; **middle:** Random Forest; **bottom:** CNN.

FIGURE 8. Summary of average performance of ML models in different splits. **Top:** in-domain testing with mixed-samples split; **middle:** in-domain testing with PRN-level split; **bottom:** cross-domain testing.

followed by CNN and Random Forest. In addition, XGBoost offered the lowest computational cost during testing and the second-lowest during training.

VI. CONCLUSION

Our paper offered a comprehensive analysis of in-domain and cross-domain spoofing detection mechanisms with GNSS post-correlation data. We compared four classical ML algorithms, namely XGBoost, Random Forest, kNN and CNN by suitably selecting eight post-correlation features and by looking at eight TEXBAT and four OAKBAT datasets. We also investigated three different approaches: in-domain testing with mixed samples, in-domain testing with PRN-level split and cross-domain testing. Compared to the other models, XGBoost outperformed the other algorithms in terms of tradeoffs between the spoofing detection

accuracy and computational efficiency. The comparison between in-domain and cross-domain testing with TEXBAT and OAKBAT datasets has not been addressed in the literature so far. This study contributes to adopting a more realistic approach to detect GNSS spoofing by exploiting per-satellite signal anomalies, as spoofing attacks typically affect only a subset of satellites. TEXBAT and OAKBAT can be used as benchmarks in assessing GNSS spoofing, but do not cover all possible spoofing scenarios. Future work will also cover other open-access scenarios, such as FGI datasets [46] and TUNI datasets [47]. The cross-domain testing showed a significant gap in performance, of about 30% compared to the in-domain testing, This also points out towards the need of using multiple datasets for training and investigating more advanced ML techniques, such as meta-learning, semi-supervised or hybrid learning techniques. Future research can also take advantage of the navigation layer to ensure more advanced, scalable, and practical solutions in the GNSS security domain.

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